Relativity and Electromagnetism Homework Assignment 3

Submission deadline: 2024 May 08 (please submit by 17:00 CEST)

[*Note:* This is the third homework assignment on the basics of electromagnetism. It contributes to 10% of the total points for final evaluation. You will be graded not only on the final result of a problem, but also on the reasoning and the intermediate steps. Please upload a (LaTeX / scanned) **pdf** copy of your answer sheet on Ufora by the deadline indicated above.]

Problem 1. The covariant electromagnetic *energy-momentum-stress tensor* is defined as

$$\mathbb{T}^{\mu\nu} \equiv \frac{1}{\mu_0}\left[F^{\mu\alpha}F^{\nu}{}_{\alpha} - \frac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}\right],$$

where $F^{\mu\nu}$ is the usual electromagnetic field tensor, defined (in our conventions) as

$$F^{\mu\nu} \equiv \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix}.$$

Show that:

$$\mathbb{T}^{00} = \frac{1}{2}\left[\epsilon_0 E^2 + \frac{1}{\mu_0}B^2\right] = u_{\rm EM}\,, \text{ the electromagnetic energy density,}$$

$$c\,\mathbb{T}^{0i} = \frac{1}{\mu_0}\epsilon^{ijk}E_jB_k = S_i\,, \text{ the Poynting vector, and}$$

$$-\mathbb{T}^{ij} = \epsilon_0\left(E_iE_j - \frac{1}{2}\delta_{ij}E^2\right) + \frac{1}{\mu_0}\left(B_iB_j - \frac{1}{2}\delta_{ij}B^2\right) = T_{ij}\,, \text{ the usual Maxwell stress tensor.}$$

Hence show that the covariant conservation equation $\partial_\mu \mathbb{T}^{\mu\nu} = 0$ implies energy as well as momentum conservation. **[3 points]**

Problem 2. The covariant form of the Lorentz force law reads

$$K^\mu = q\,\eta_\nu\,F^{\mu\nu}\,,$$

where q is the charge of the particle, $\eta^\mu \equiv (\gamma c, \gamma \boldsymbol{u})$ its proper velocity, $F^{\mu\nu}$ the field tensor, and K^μ is the Minkowski force defined as $K^\mu \equiv \dfrac{dp^\mu}{d\tau}$, with proper momentum $p^\mu \equiv m\eta^\mu$.

a. Show that the spatial components of this covariant expression give the usual force law

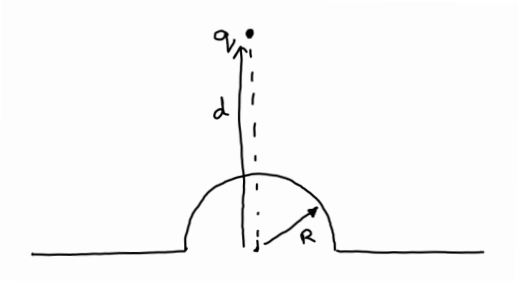
$$\boldsymbol{F} = q\,[\boldsymbol{E} + (\boldsymbol{u}\times\boldsymbol{B})]\,.$$

b. Show that the zeroth component gives

$$K^0 = \frac{\gamma q}{c}\,\boldsymbol{u}\cdot\boldsymbol{E}\,.$$

Note: $K^0 \equiv \dfrac{1}{c}\dfrac{dE}{d\tau} = \dfrac{\gamma}{c}\dfrac{dE}{dt}$. This result shows that the power delivered to the particle is the force $q\boldsymbol{E}$ times the velocity $\boldsymbol{u}$. **[1.5 points]**

Problem 3. A conducting surface of the form of an infinite flat plate with a hemisphere of radius R embossed on top of it (see figure) is placed along the xy plane. A point charge q is placed on the z axis at a distance d from the centre of the spherical surface.



a. Calculate the electrostatic potential $V(x, y, z)$ or equivalently $V(r, \theta, \phi)$ at an arbitrary point above the surface. Use the method of images and feel free to use the result from Problem 3.2 of the Griffiths textbook and any associated expressions.

b. Write down an expression for the surface charge density on the conducting surface in terms of $V(x, y, z)$ and/or $V(r, \theta, \phi)$. **Note:** *You do not need to evaluate the expression. Just write it down!* You may want to use different coordinate systems (Cartesian, shperical) for different parts of the surface.

c. What is the total induced charge on the conducting surface? **Note:** You are *not* expected to perform an integration over the surface charge density for this problem.

d. Sketch the approximate electric field lines assuming that the charge q is positive.

[2 points]

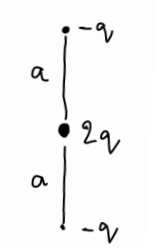
Problem 4. A thin spherical shell of radius R is maintained at an electrostatic potential with an angular dependence

$$V(\theta) = V_0 \cos^2 \theta \, .$$

a. Calculate the radial dependence of the potential inside and outside the sphere.

b. What is the electrostatic dipole moment of the system? **[2 points]**

Problem 5. Three point charges $2q$, $-q$, and $-q$ are placed on the z axis at $z = 0$, $z = a$, and $z = -a$ respectively (see figure).



Calculate the monopole, dipole, and quadrupole moments for this charge configuration.

[1.5 points]